HANKEL OPERATORS FROM A^1_ω TO L^1_ω WITH LOGARITHMIC BMO SYMBOLS

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ABSTRACT. Let ω be a radial weight on the unit disc which is both doubling and reverse doubling, and let $H^\omega_f = (I - P_\omega)M_f$ be the Hankel operator with symbol f on the weighted Bergman space A^1_ω. Duan, Guo, Hong and Yan asked whether H^ω_f is bounded from A^1_ω to L^1_ω for every symbol f of logarithmic bounded mean oscillation, and noted that this was open even for the classical Bergman space. We show that the answer is yes, without any boundedness assumption on the symbol and also for the largest space, defined by means of L^1 oscillation. The proof combines a smooth decomposition of the symbol with a non-canonical solution of a $\overline{\partial}$-equation, which is made canonical by the weighted Bergman projection.

1. INTRODUCTION

Let $\mathbb{D}$ be the unit disc and $\mathrm{d}A(z) = \mathrm{d}x\,\mathrm{d}y/\pi$ the normalized area measure. For a radial weight ω on $\mathbb{D}$ we write

$$\widehat{\omega}(r) = \int_r^1 \omega(s)\,\mathrm{d}s, \qquad \widetilde{\omega}(z) = \frac{\widehat{\omega}(|z|)}{1-|z|},$$

and we assume throughout that $\widehat{\omega}(r) > 0$ for $0 \le r < 1$. The weight ω is *doubling*, $\omega \in \widehat{\mathcal{D}}$, if there is $C \ge 1$ such that $\widehat{\omega}(r) \le C\widehat{\omega}((1+r)/2)$ for $0 \le r < 1$, and *reverse doubling*, $\omega \in \check{\mathcal{D}}$, if there are $K > 1$ and $C > 1$ such that $\widehat{\omega}(r) \ge C\widehat{\omega}(1-(1-r)/K)$ for $0 \le r < 1$. We set $\mathcal{D} = \widehat{\mathcal{D}} \cap \check{\mathcal{D}}$. The standard weights $(\alpha+1)(1-|z|^2)^\alpha$, $\alpha > -1$, belong to $\mathcal{D}$.

For $1 \le p < \infty$ let $L^p_\omega = L^p(\mathbb{D}, \omega\,\mathrm{d}A)$, and let A^p_ω be the subspace of analytic functions, the weighted Bergman space with norm

$$\|g\|^p_{A^p_\omega} = \int_{\mathbb{D}} |g(z)|^p \omega(z)\,\mathrm{d}A(z).$$

The reproducing kernel of A^2_ω is denoted by B^ω_z, and P_ω is the orthogonal projection of L^2_ω onto A^2_ω,

$$P_\omega u(z) = \int_{\mathbb{D}} u(\zeta)\overline{B^\omega_z(\zeta)}\omega(\zeta)\,\mathrm{d}A(\zeta).$$

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For a symbol $f \in L^2_\omega$, Duan, Guo, Hong and Yan [2] consider the Hankel operator

$$H^\omega_f g = (I - P_\omega)(fg), \qquad g \in H^\infty,$$

and ask when it extends to a bounded operator from A^1_ω to L^1_ω.

The relevant symbol classes are defined in the Bergman metric $\beta(z,w) = \operatorname{arctanh}|\varphi_z(w)|$, where $\varphi_z(w) = (z-w)/(1-\overline{z}w)$. Let $D(z,r)$ be the Bergman disc of centre z and radius r. For $\omega \in \mathcal{D}$ there is $r_0 = r_0(\omega) > 0$ such that [2, (1.2)]

$$\omega(D(z,r)) \asymp \widehat{\omega}(|z|)(1-|z|), \qquad z \in \mathbb{D},\ r \geq r_0, \tag{1}$$

with constants depending on ω and r. For $r \geq r_0$ and $f \in L^1_\omega$ we write $\widehat{f}_{r,\omega}(z)$ for the mean of f over $D(z,r)$ with respect to $\omega\,\mathrm{d}A$, and

$$\mathrm{MO}^p_{\omega,r}(f)(z) = \left(\frac{1}{\omega(D(z,r))} \int_{D(z,r)} \left|f(\zeta) - \widehat{f}_{r,\omega}(z)\right|^p \omega(\zeta)\,\mathrm{d}A(\zeta) \right)^{1/p}.$$

Following [2, Section 3.4], the space $\mathrm{BMO}^p_{\omega,r,\log}$ consists of the $f \in L^p_\omega$ with

$$\|f\|_{\mathrm{BMO}^p_{\omega,r,\log}} = \sup_{z\in\mathbb{D}} \log\frac{e}{1-|z|}\,\mathrm{MO}^p_{\omega,r}(f)(z) < \infty.$$

Duan, Guo, Hong and Yan state that this space does not depend on $r \geq r_0$, and denote it by $\mathrm{BMO}^p_{\omega,\log}$.

Let $\mathcal{LB}$ denote the logarithmic Bloch space of analytic functions f in $\mathbb{D}$ with

$$\|f\|_{\mathcal{LB}} = \sup_{z\in\mathbb{D}} |f'(z)|(1-|z|^2)\log\frac{e}{1-|z|} < \infty.$$

In the classical Bergman space A^1, Attele [1] characterized the bounded Hankel operators $H_{\overline{f}}\colon A^1 \to L^1$ with analytic f in terms of $\mathcal{LB}$; see [2, Section 1]. For general symbols, Taskinen and Virtanen [6, Corollary 3.5] showed that H_f is bounded from A^1 to L^1 if $f = f_1 + f_2$, where f_1 is bounded and has logarithmically bounded oscillation and f_2 has logarithmically bounded averages. They asked whether the boundedness of f_1 can be dropped, that is, whether H_f is bounded for every f of logarithmic bounded mean oscillation in the unweighted L^1 sense; see [6, Remark 3.6] and [5, Section 5]. Duan, Guo, Hong and Yan proved the analogue of the Taskinen–Virtanen result for $\omega \in \mathcal{D}$, with f_2 of logarithmically bounded L^p_ω averages for some $p > 1$ [2, Proposition 22], and they extended Attele's theorem: for $\omega \in \mathcal{D}$ and $f \in A^2_\omega$, the operator $H^\omega_{\overline{f}}$ is bounded from A^1_ω to L^1_ω if and only if $f \in \mathcal{LB}$ [2, Theorem 5]. Both positive results for general symbols require the oscillating part of the symbol to be bounded. The final sentence before the references of [2], at the end of Section 4, reads:

> For $1 < p < \infty$, the problem whether $H^\omega_f : A^1_\omega \to L^1_\omega$ is bounded for every $f \in \mathrm{BMO}^p_{\omega,\log}$ remains open even in the classical Bergman space.

The main purpose of the present paper is to answer this question in the affirmative. Since $\mathrm{MO}^1_{\omega,r}(f) \le \mathrm{MO}^p_{\omega,r}(f)$ by Hölder's inequality, we have $\mathrm{BMO}^p_{\omega,r,\log} \subset \mathrm{BMO}^1_{\omega,r,\log}$ for every $p \ge 1$, and we state the result for the largest of these spaces. For symbols that are not in L^2_ω, the projection in the definition of H^ω_f is understood as the integral above; we show that it converges absolutely.

Theorem 1. *Suppose that $\omega \in \mathcal{D}$, that $1 \le p < \infty$, that $r > r_0$, and that $f \in \mathrm{BMO}^p_{\omega,r,\log}$. Then $fg \in L^1_\omega$ for every $g \in H^\infty$, the integral defining $P_\omega(fg)$ converges absolutely, and*

$$\|H^\omega_f g\|_{L^1_\omega} \le C\|f\|_{\mathrm{BMO}^p_{\omega,r,\log}}\|g\|_{A^1_\omega}, \qquad g \in H^\infty,$$

where C depends only on ω and r. If $f \in L^2_\omega$, then $H^\omega_f g = (I - P_\omega)(fg)$ in the sense of L^2_ω. Consequently H^ω_f extends to a bounded operator from A^1_ω to L^1_ω.

To the best of our knowledge, Theorem 1 is new also for the unweighted Bergman space A^1. In comparison with [6, Corollary 3.5] and [2, Proposition 22], no part of the symbol is assumed to be bounded. For $\omega \equiv 1$ and $p = 1$, the space $\mathrm{BMO}^1_{\omega,r,\log}$ is the space $\mathrm{BMO}^1_{r,\log}$ of Taskinen and Virtanen, and (1) holds for every $r > 0$; hence Theorem 1 also answers the question in [6, Remark 3.6] and [5, Section 5] in the affirmative. The restriction $r > r_0$ is discussed in Section 6. It excludes at most one radius, the smallest one for which (1) holds, if there is a smallest one; if the independence of $r \ge r_0$ stated in [2] is granted, Theorem 1 covers all of $\mathrm{BMO}^p_{\omega,\log}$.

Corollary 1. *Suppose that $\omega \in \mathcal{D}$ and $f \in \mathcal{LB}$. Then $H^\omega_{\overline{f}}$ is bounded from A^1_ω to L^1_ω.*

Corollary 1 is the sufficiency part of [2, Theorem 5], and for $\omega \equiv 1$ of Attele's theorem. The proof given in [2] passes through an auxiliary integral operator from L^∞ to the Bloch space; here it follows from Theorem 1 and the observation that $\overline{f} \in \mathrm{BMO}^1_{\omega,r,\log}$ for $f \in \mathcal{LB}$.

The idea of the proof. Let us briefly describe our approach. As in the Hilbert space theory [7], we split $f = F + h$, where F is smooth with $|\nabla F(z)| \lesssim 1/((1-|z|)\log(e/(1-|z|)))$ and h is small on average. The part h is handled by a Carleson-type estimate and the logarithmic L^1 bound for P_ω due to Peláez and Rättyä [4]. For F we use that $H^\omega_F g$ is the solution of $\overline{\partial} u = g\overline{\partial}F$ which is orthogonal to A^2_ω. There is no explicit formula for this canonical solution when ω is a general weight in $\mathcal{D}$. Instead we write down an explicit non-canonical solution,

$$u(z) = \int_{\mathbb{D}} \frac{(1-|\zeta|^2)^N}{(z-\zeta)(1-z\overline{\zeta})^N} g(\zeta)\overline{\partial}F(\zeta)\,\mathrm{d}A(\zeta),$$

and observe that $H^\omega_F g = (I - P_\omega)u$. The Peláez–Rättyä bound costs a factor $\log(e/(1-|z|))$, so it suffices to estimate u in L^1_ω with this extra logarithmic

weight. Near the diagonal the kernel is the Cauchy kernel, and its integral over a disc of radius comparable to $1-|\zeta|$ cancels the factor $1/(1-|\zeta|)$ in $\overline{\partial}F$; the logarithm in the gradient bound cancels the logarithm lost by P_ω. Away from the diagonal the factor $((1-|\zeta|^2)/(1-z\overline{\zeta}))^N$ makes the kernel decay faster than the doubling condition allows $\widehat{\omega}$ to grow. This is the only place where the size of N matters. In contrast to the proof of [2, Proposition 22], which estimates the oscillating part of H^ω_f through the Toeplitz operator T^ω_f and the bound $\|fg\|_{L^1_\omega} \leq \|f\|_\infty \|g\|_{A^1_\omega}$, we never pass through T^ω_f.

Organization. The present paper is organized into five additional sections. In Section 2 we collect the facts on doubling weights, Bergman discs and the projection P_ω that we use. In Section 3 we decompose a symbol of logarithmic mean oscillation. In Section 4 we solve the $\overline{\partial}$-equation with a logarithmic L^1 estimate. Section 5 contains the proofs of Theorem 1 and Corollary 1, and Section 6 contains concluding remarks.

2. Preliminaries

Throughout, $\omega \in \mathcal{D}$ is fixed, and

$$\delta(z) = 1-|z|, \qquad \ell(z) = \log \frac{e}{1-|z|},$$

also for real $0 \leq z < 1$. Note that $\ell \geq 1$, and that $x \mapsto x\log(e/x)$ is increasing on $0 < x \leq 1$, so that $\delta\ell$ increases when δ does. We write $X \lesssim Y$ if $X \leq CY$ for a constant C that depends only on ω and on the fixed radii and integers in the statement at hand, and $X \asymp Y$ if $X \lesssim Y \lesssim X$.

Doubling weights. By [3, Lemma 2.1], see also [2, Lemma A, part (ii)], there are $C \geq 1$ and $b > 0$ such that

$$\widehat{\omega}(r) \leq C\left(\frac{1-r}{1-t}\right)^b \widehat{\omega}(t), \qquad 0 \leq r \leq t < 1. \tag{2}$$

Reverse doubling gives, conversely, $C \geq 1$ and $a > 0$ such that

$$\widehat{\omega}(t) \leq C\left(\frac{1-t}{1-r}\right)^a \widehat{\omega}(r), \qquad 0 \leq r \leq t < 1; \tag{3}$$

see [2, Lemma B]. Indeed, if $1-t \leq K^{-k}(1-r)$, then k applications of the definition give $\widehat{\omega}(t) \leq C^{-k}\widehat{\omega}(r)$, and (3) follows with $a = \log C/\log K$. We write b and a for the exponents called β and α in [2], since β denotes the Bergman metric. The exponent b measures how fast $\widehat{\omega}$ may grow away from the boundary, and a how fast it must decay towards it.

The next lemma collects the integral estimates that we need. Part (iii) is the form in which all weighted area estimates below are made: we only ever integrate ω over regions that are full in the radial direction up to the boundary, so that only the tails $\widehat{\omega}$ enter.

Lemma 1. *Suppose that $\omega \in \mathcal{D}$. Then the following hold.*

(1) *For $m = 1, 2$ and $0 \leq r < 1$, $\int_r^1 \ell(s)^m \omega(s)\,\mathrm{d}s \lesssim \ell(r)^m \widehat{\omega}(r)$.*

(2) *For* $0 \le r < 1$, $\int_r^1 \widehat{\omega}(s)(1-s)^{-1}\,\mathrm{d}s \asymp \widehat{\omega}(r)$.
(3) *Suppose that* $t > 0$, $c > 0$ *and* $\theta \in \mathbb{R}$, *and that* $E \subset \mathbb{D}$ *is contained in the polar box* $\{\rho e^{i\vartheta} : \rho > 1-t,\ |\vartheta - \theta| < ct\}$. *Then, with* $r_t = \max(0, 1-t)$,

$$\int_E \omega\,\mathrm{d}A \le \frac{2ct}{\pi}\widehat{\omega}(r_t), \qquad \int_E \ell\omega\,\mathrm{d}A \le \frac{2ct}{\pi}\int_{r_t}^1 \ell(s)\omega(s)\,\mathrm{d}s \lesssim c\,t\,\ell(r_t)\widehat{\omega}(r_t).$$

Proof. (i) Split $[r, 1)$ into the intervals $I_k = [1-2^{-k}(1-r), 1-2^{-k-1}(1-r))$, $k \ge 0$. On I_k we have $\ell(s) \le \ell(r) + (k+1)\log 2 \le (k+2)\ell(r)$, since $\ell(r) \ge 1$, and (3) gives

$$\int_{I_k} \omega(s)\,\mathrm{d}s \le \widehat{\omega}(1-2^{-k}(1-r)) \lesssim 2^{-ka}\widehat{\omega}(r).$$

Hence the integral is at most a constant times $\ell(r)^m\widehat{\omega}(r)\sum_{k\ge 0}(k+2)^m 2^{-ka}$.

(ii) For the upper bound we use the same intervals. On I_k, $\widehat{\omega}(s) \lesssim 2^{-ka}\widehat{\omega}(r)$ by (3), and $\int_{I_k}(1-s)^{-1}\,\mathrm{d}s = \log 2$. For the lower bound we integrate only over $[r, (1+r)/2]$, where $\widehat{\omega}(s) \ge \widehat{\omega}((1+r)/2) \gtrsim \widehat{\omega}(r)$ by (2).

(iii) In polar coordinates $\mathrm{d}A = \pi^{-1}\rho\,\mathrm{d}\rho\,\mathrm{d}\vartheta$, and $\rho \le 1$. The angular integral is at most $2ct$, and the radial one is taken over $(r_t, 1)$. The last inequality is (i). □

The next lemma is [2, Lemma F] for $p = 1$; we include the short proof, which only uses the tails of ω.

Lemma 2. *Suppose that* $\omega \in \mathcal{D}$. *Then* $\|g\|_{A^1_{\widetilde{\omega}}} \asymp \|g\|_{A^1_\omega}$ *for every analytic function* g *in* $\mathbb{D}$.

Proof. Let $M(s)$ be the mean of $|g|$ over the circle of radius s. Then

$$\|g\|_{A^1_\omega} = \int_0^1 M(s)\,\mathrm{d}\mu(s), \qquad \|g\|_{A^1_{\widetilde{\omega}}} = \int_0^1 M(s)\,\mathrm{d}\nu(s),$$

where $\mathrm{d}\mu(s) = 2s\omega(s)\,\mathrm{d}s$ and $\mathrm{d}\nu(s) = 2s\widetilde{\omega}(s)\,\mathrm{d}s$. We claim that $\mu([t,1)) \asymp \nu([t,1)) \asymp \widehat{\omega}(t)$ for $0 \le t < 1$. For $t \ge 1/2$ the factor $2s$ lies between 1 and 2, and the claim follows from the definition of $\widehat{\omega}$ and from Lemma 1 (ii). For $t < 1/2$ both tails lie between their values at $t = 1/2$ and at $t = 0$, and these are comparable to $\widehat{\omega}(1/2)$ and to $\widehat{\omega}(0)$, which are comparable by (2). Since $|g|$ is subharmonic, M is nondecreasing. Hence every superlevel set $\{M > \lambda\}$ is an interval with right endpoint 1, and the layer cake formula $\int_0^1 M\,\mathrm{d}\mu = \int_0^\infty \mu(\{M > \lambda\})\,\mathrm{d}\lambda$, together with the same formula for ν, gives the assertion. □

Bergman discs. We record three elementary facts. First, if $\beta(z, w) < s$, then

$$\frac{\delta(w)}{4\cosh^2 s} \le \delta(z) \le 4\cosh^2 s\,\delta(w), \qquad \ell(z) \lesssim \ell(w), \qquad \widehat{\omega}(|z|) \asymp \widehat{\omega}(|w|). \tag{4}$$

Indeed, $1 - |\varphi_w(z)|^2 = (1-|w|^2)(1-|z|^2)/|1-\overline{w}z|^2$. This quantity exceeds $1 - \tanh^2 s = 1/\cosh^2 s$, and, since $|1-\overline{w}z| \ge \delta(w)$, it is at most $4\delta(z)/\delta(w)$.

The second inequality follows by symmetry, the bound for ℓ follows by taking logarithms, and the last estimate follows from the monotonicity of $\widehat{\omega}$ and (2). In view of (1), the masses $\omega(D(z,R))$ and $\omega(D(w,R'))$ are therefore comparable when $\beta(z,w) < s$ and $R, R' \geq r_0$.

Second, if $|u - z| < \delta(z)/2$, then $|\varphi_z(u)| \leq 1/2$, because $|1 - \overline{z}u| \geq \delta(z)$. Hence the Euclidean disc of centre z and radius $\delta(z)/2$ is contained in $D(z,1)$.

Third, we use a lattice. Let $\{a_j\}$ be a maximal subset of $\mathbb{D}$ with $\beta(a_j, a_k) \geq 1$ for $j \neq k$. By maximality, the discs $D(a_j, 1)$ cover $\mathbb{D}$. The discs $D(a_j, 1/2)$ are disjoint, and the Möbius invariant measure $\mathrm{d}\lambda(z) = (1-|z|^2)^{-2}\,\mathrm{d}A(z)$ satisfies $\lambda(D(z,s)) = \lambda(D(0,s)) = \sinh^2 s$. Comparing measures, we find that

$$\#\{j : \beta(z, a_j) < 4\} \leq K_0 = \frac{\sinh^2(9/2)}{\sinh^2(1/2)}, \qquad z \in \mathbb{D}. \tag{5}$$

The Bergman projection on L^1. The Bergman projection is not bounded on L^1_ω, but it loses only a logarithm. This is the source of the logarithm in the symbol class.

Lemma 3. *Suppose that $\omega \in \mathcal{D}$, and that u is measurable with $\int_{\mathbb{D}} \ell|u|\omega\,\mathrm{d}A < \infty$. Then the integral defining $P_\omega u(z)$ converges absolutely for every $z \in \mathbb{D}$, and*

$$\|P_\omega u\|_{L^1_\omega} \lesssim \int_{\mathbb{D}} \ell|u|\omega\,\mathrm{d}A.$$

Proof. Since $B^\omega_z(\zeta) = \sum_{n\geq 0}(\overline{z}\zeta)^n/(2\omega_{2n+1})$ with $\omega_x = \int_0^1 s^x\omega(s)\,\mathrm{d}s \geq \varrho^x\widehat{\omega}(\varrho)$ for every $0 < \varrho < 1$, the kernel B^ω_z is bounded on $\mathbb{D}$ for fixed z; this gives the absolute convergence. By Tonelli's theorem and the symmetry $|B^\omega_z(\zeta)| = |B^\omega_\zeta(z)|$,

$$\|P_\omega u\|_{L^1_\omega} \leq \int_{\mathbb{D}} |u(\zeta)|\|B^\omega_\zeta\|_{A^1_\omega}\omega(\zeta)\,\mathrm{d}A(\zeta).$$

By [4, Theorem 1(ii)] with $p = 1$, $N = 0$ and $v = \omega$,

$$\|B^\omega_\zeta\|_{A^1_\omega} \asymp \int_0^{|\zeta|} \frac{\mathrm{d}t}{1-t} = \log\frac{1}{1-|\zeta|}, \qquad |\zeta| \to 1^-,$$

see also [2, Lemma D]. For $|\zeta| \leq \varrho < 1$ we have $\|B^\omega_\zeta\|_{A^1_\omega} \leq \omega(\mathbb{D})^{1/2}B^\omega_\varrho(\varrho)^{1/2}$. Hence $\|B^\omega_\zeta\|_{A^1_\omega} \lesssim \ell(\zeta)$ for every $\zeta \in \mathbb{D}$. □

3. A decomposition of the symbol

We first show that the logarithmic oscillation condition at one radius $s > r_0$ implies the same condition at every larger radius. The point is that two overlapping Bergman discs share a disc of radius at least r_0, which has comparable weighted mass by (1), so that their weighted means differ by $O(1/\ell)$.

Lemma 4. *Suppose that $\omega \in \mathcal{D}$, that $r_0 < s < r$, and that $f \in L^1_\omega$. Then*

$$\sup_{z\in\mathbb{D}} \ell(z)\mathrm{MO}^1_{\omega,r}(f)(z) \lesssim \sup_{z\in\mathbb{D}} \ell(z)\mathrm{MO}^1_{\omega,s}(f)(z).$$

Proof. Let M be the right-hand side and $\tau = 2(s - r_0)$. Suppose first that $\beta(x, y) \le \tau$, and let m be the midpoint of the geodesic from x to y. Then $D(m, r_0) \subset D(x, s) \cap D(y, s)$, by the triangle inequality. Hence

$$\begin{aligned} |\widehat{f}_{s,\omega}(x) - \widehat{f}_{r_0,\omega}(m)| &\le \frac{1}{\omega(D(m, r_0))} \int_{D(m,r_0)} |f - \widehat{f}_{s,\omega}(x)|\omega \,\mathrm{d}A \\ &\le \frac{\omega(D(x, s))}{\omega(D(m, r_0))}\mathrm{MO}^1_{\omega,s}(f)(x), \end{aligned}$$

which is $\lesssim M/\ell(x)$ by (1) and (4). The same holds with y in place of x, and so $|\widehat{f}_{s,\omega}(x) - \widehat{f}_{s,\omega}(y)| \lesssim M/\ell(x)$.

Next we fix z. Since the closure of $D(0, r)$ is compact, it is covered by finitely many discs $D(y_i, s)$ with $y_i \in D(0, r)$, $1 \le i \le n$. Since φ_z is an isometry for β and maps 0 to z, the discs $D(x_i, s)$ with $x_i = \varphi_z(y_i)$ cover $D(z, r)$, and $x_i \in D(z, r)$. Joining z to x_i by a chain of at most $\lceil r/\tau \rceil$ points on the geodesic, with consecutive distances at most τ, and using that $\ell \asymp \ell(z)$ on $D(z, r)$ by (4), we find that $d_i = |\widehat{f}_{s,\omega}(x_i) - \widehat{f}_{s,\omega}(z)| \lesssim M/\ell(z)$. Consequently

$$\begin{aligned} \int_{D(z,r)} |f - \widehat{f}_{s,\omega}(z)|\omega \,\mathrm{d}A &\le \sum_{i=1}^{n} \omega(D(x_i, s)) \left(\mathrm{MO}^1_{\omega,s}(f)(x_i) + d_i\right) \\ &\lesssim \frac{\omega(D(z, r))M}{\ell(z)}, \end{aligned}$$

where we used (1) and (4) once more to compare $\omega(D(x_i, s))$ with $\omega(D(z, r))$. Finally, replacing $\widehat{f}_{r,\omega}(z)$ by any constant c at most doubles the oscillation, since $|\widehat{f}_{r,\omega}(z) - c|$ is at most the mean of $|f - c|$ over $D(z, r)$. □

The next lemma is our version of the decomposition into a part of bounded oscillation and a part of bounded averages; compare [7] in the Hilbert space setting, [6, Proposition 2.1] for the unweighted logarithmic spaces, and [2, Proposition 13], whose proof is omitted there. The smooth part F is obtained by gluing the local means of f with a partition of unity adapted to the Bergman metric. Neighbouring means differ by $O(1/\ell)$, and the partition functions vary on the scale δ, which gives the gradient bound. What we need from the remainder h is not a pointwise bound but its effect against analytic functions, with the extra weight ℓ.

Lemma 5. *Suppose that $\omega \in \mathcal{D}$, that $\varrho \ge r_0 + 3$, that $f \in L^1_\omega$, and that*

$$M = \sup_{z\in\mathbb{D}} \ell(z)\mathrm{MO}^1_{\omega,\varrho}(f)(z) < \infty.$$

Then $f = F + h$, where F is a smooth function on $\mathbb{D}$ with

$$|\partial_x F(z)| + |\partial_y F(z)| \lesssim \frac{M}{\delta(z)\ell(z)}, \qquad z \in \mathbb{D}, \tag{6}$$

and

$$\int_{\mathbb{D}} \ell |hg| \omega \, \mathrm{d}A \lesssim M \|g\|_{A^1_\omega} \tag{7}$$

for every analytic function g in $\mathbb{D}$.

Proof. Let $R = \varrho - 3 \geq r_0$, and let $\{a_j\}$ be the lattice from Section 2. Fix a smooth function χ on $\mathbb{R}$ with $0 \leq \chi \leq 1$, $\chi(x) = 1$ for $x \leq \tanh^2 1$ and $\chi(x) = 0$ for $x \geq \tanh^2 2$. Put $\psi_j(z) = \chi(|\varphi_{a_j}(z)|^2)$. Then ψ_j is smooth, equals 1 on $D(a_j, 1)$ and vanishes outside $D(a_j, 2)$. Hence $1 \leq \Psi = \sum_k \psi_k \leq K_0$ by (5), and $\eta_j = \psi_j / \Psi$ is a smooth partition of unity in which at most K_0 terms are nonzero at each point. Since $|\nabla |\varphi|^2| = 2|\varphi||\varphi'|$ for analytic φ and

$$|\varphi_a'(z)| = \frac{1 - |a|^2}{|1 - \overline{a} z|^2} = \frac{1 - |\varphi_a(z)|^2}{1 - |z|^2} \leq \frac{1}{\delta(z)},$$

we have $|\nabla \psi_j| \lesssim 1/\delta$, and therefore $|\nabla \eta_j| \lesssim 1/\delta$.

Let $c_j = \widehat{f}_{R,\omega}(a_j)$ and $F = \sum_j \eta_j c_j$. This is a locally finite sum of smooth functions. Fix z. If $\eta_j(z) \neq 0$, then $\beta(z, a_j) < 2$, so that $D(a_j, R) \subset D(z, \varrho)$, and

$$|c_j - \widehat{f}_{\varrho,\omega}(z)| \leq \frac{\omega(D(z,\varrho))}{\omega(D(a_j, R))} \mathrm{MO}^1_{\omega,\varrho}(f)(z) \lesssim \frac{M}{\ell(z)}$$

by (1) and (4). Since $\sum_j \eta_j = 1$, we have $\sum_j \nabla \eta_j = 0$, and so

$$\nabla F(z) = \sum_j \nabla \eta_j(z) \left(c_j - \widehat{f}_{\varrho,\omega}(z) \right).$$

At most K_0 terms are nonzero, and (6) follows.

Let $h = f - F = \sum_k \eta_k (f - c_k)$. Fix j. If $z \in D(a_j, 1)$ and $\eta_k(z) \neq 0$, then $\beta(a_j, a_k) < 3$, and so $D(a_k, R) \subset D(a_j, \varrho)$. Arguing as above with a_j in place of z, we get $|c_k - \widehat{f}_{\varrho,\omega}(a_j)| \lesssim M/\ell(a_j)$. Hence, by (5),

$$\begin{aligned} \int_{D(a_j,1)} |h| \omega \, \mathrm{d}A &\leq \sum_{k : \beta(a_j,a_k) < 3} \int_{D(a_j,\varrho)} \left(|f - \widehat{f}_{\varrho,\omega}(a_j)| + |\widehat{f}_{\varrho,\omega}(a_j) - c_k| \right) \omega \, \mathrm{d}A \\ &\lesssim \frac{M \delta(a_j) \widehat{\omega}(|a_j|)}{\ell(a_j)}, \end{aligned}$$

where we used (1) in the last step. By the second fact in Section 2, the Euclidean disc of centre $u \in D(a_j, 1)$ and radius $\delta(u)/2$ lies in $D(a_j, 2)$. By the subharmonicity of $|g|$ and (4),

$$\sup_{D(a_j,1)} |g| \leq \sup_{u \in D(a_j,1)} \frac{4}{\delta(u)^2} \int_{|v-u| < \delta(u)/2} |g(v)| \, \mathrm{d}A(v) \lesssim \frac{1}{\delta(a_j)^2} \int_{D(a_j,2)} |g| \, \mathrm{d}A.$$

Since $\ell \lesssim \ell(a_j)$ on $D(a_j, 1)$, we find that

$$\int_{D(a_j,1)} \ell |hg| \omega \, \mathrm{d}A \lesssim M \frac{\widehat{\omega}(|a_j|)}{\delta(a_j)} \int_{D(a_j,2)} |g| \, \mathrm{d}A \lesssim M \int_{D(a_j,2)} |g| \widetilde{\omega} \, \mathrm{d}A,$$

by (4) once more. We sum over j. The discs $D(a_j, 1)$ cover $\mathbb{D}$, and the discs $D(a_j, 2)$ overlap at most K_0 times by (5). Hence $\int_{\mathbb{D}} \ell |hg| \omega \, \mathrm{d}A \lesssim M \|g\|_{A^1_{\widetilde{\omega}}}$, and Lemma 2 gives (7). □

In the notation of [2], Lemma 5 is a quantitative form of the inclusion $\mathrm{BMO}^1_{\omega,\log} \subset \mathrm{BO}_{\log} + \mathrm{BA}^1_{\omega,\log}$. Indeed, $F \in \mathrm{BO}_{\log}$ by (6), and the bound for $\int_{D(a_j,1)} |h| \omega \, \mathrm{d}A$ in the proof, summed over the boundedly many a_j with $D(a_j, 1) \cap D(z, r) \neq \emptyset$, shows that $h \in \mathrm{BA}^1_{\omega,r,\log}$ for $r \geq r_0$. Estimate (7) is the form in which we use h.

4. A $\overline{\partial}$-equation with a logarithmic L^1 estimate

Let $\overline{\partial} = (\partial_x + i\partial_y)/2$. For a positive integer N we set

$$S_N(z, \zeta) = \frac{(1 - |\zeta|^2)^N}{(z - \zeta)(1 - z\overline{\zeta})^N}, \qquad z, \zeta \in \mathbb{D},\ z \neq \zeta.$$

As a function of z, this kernel has a simple pole at ζ with residue 1, and is otherwise analytic in $\mathbb{D}$. The factor $((1 - |\zeta|^2)/(1 - z\overline{\zeta}))^N$ equals 1 at $z = \zeta$, is bounded by 2^N, and decays like $(\delta(\zeta)/|1 - z\overline{\zeta}|)^N$. Since $1 - |\zeta|^2 \leq 2\delta(\zeta) \leq 2|1 - z\overline{\zeta}|$, we have

$$|S_N(z, \zeta)| \leq \frac{2^N}{|z - \zeta|}, \qquad z, \zeta \in \mathbb{D}. \tag{8}$$

The decay is needed against the weight $\ell\omega$, through the following lemma. It is a logarithmic version of the familiar estimate $\int_{\mathbb{D}} \omega(z) |1 - \overline{\zeta} z|^{-\lambda - 1} \, \mathrm{d}A(z) \asymp \widehat{\omega}(|\zeta|) \delta(\zeta)^{-\lambda}$ for suitable λ; see [2, Lemma A, part (iii)].

Lemma 6. *Suppose that $\omega \in \mathcal{D}$, and that $N > b$ is an integer, where b is the exponent in* (2)*. Then*

$$\int_{\mathbb{D}} \frac{\ell(z)\omega(z)}{|1 - z\overline{\zeta}|^{N+1}} \, \mathrm{d}A(z) \lesssim \frac{\ell(\zeta)\widehat{\omega}(|\zeta|)}{\delta(\zeta)^N}, \qquad \zeta \in \mathbb{D}.$$

Proof. Fix ζ and put $t_j = 2^j \delta(\zeta)$. Since $\delta(\zeta) \leq |1 - z\overline{\zeta}| < 2$, the shells $S_j = \{z : t_{j-1} \leq |1 - z\overline{\zeta}| < t_j\}$, $j \geq 1$, cover $\mathbb{D}$. Let $z \in S_j$. Then $\delta(z) \leq |1 - z\overline{\zeta}| < t_j$. Write $z\overline{\zeta} = \sigma e^{i\psi}$ with $\sigma \geq 0$ and $|\psi| \leq \pi$, so that $|1 - z\overline{\zeta}|^2 = (1 - \sigma)^2 + 4\sigma \sin^2(\psi/2)$. If $\sigma \geq 1/4$, then $|\psi|/\pi \leq |\sin(\psi/2)| \leq |1 - z\overline{\zeta}| < t_j$. If $\sigma < 1/4$, then $t_j > 3/4$ and $|\psi| \leq \pi < 5t_j$. In either case S_j is contained in a polar box as in Lemma 1 (iii), with $t = t_j$, $c = 5$ and $\theta = \arg \zeta$. Let $r_j = \max(0, 1 - t_j)$. Since $1 - r_j \geq \delta(\zeta)$, we have $\ell(r_j) \leq \ell(\zeta)$. Since $(1 - r_j)/\delta(\zeta) \leq 2^j$, (2) gives $\widehat{\omega}(r_j) \lesssim 2^{jb} \widehat{\omega}(|\zeta|)$. Hence Lemma 1 (iii) yields

$$\int_{S_j} \frac{\ell\omega}{|1 - z\overline{\zeta}|^{N+1}} \, \mathrm{d}A(z) \lesssim t_{j-1}^{-N-1} t_j \ell(\zeta) 2^{jb} \widehat{\omega}(|\zeta|) \lesssim 2^{-j(N-b)} \frac{\ell(\zeta)\widehat{\omega}(|\zeta|)}{\delta(\zeta)^N}.$$

The sum over $j \geq 1$ converges because $N > b$. □

We can now estimate the solution operator. The hypothesis on v below is what (6) gives for $v = g\overline{\partial}F$.

Lemma 7. *Suppose that $\omega \in \mathcal{D}$, that $N > b$ is an integer, that g is analytic in $\mathbb{D}$, and that v is measurable with $|v| \le |g|/(\delta\ell)$ on $\mathbb{D}$. Then*

$$\int_{\mathbb{D}} \ell(z)\omega(z) \int_{\mathbb{D}} |S_N(z,\zeta)v(\zeta)|\,\mathrm{d}A(\zeta)\,\mathrm{d}A(z) \lesssim \|g\|_{A^1_\omega}.$$

Proof. We split the double integral into the far region $|z-\zeta| \ge \delta(\zeta)/4$ and the near region $|z-\zeta| < \delta(\zeta)/4$.

In the far region, $|1-z\overline{\zeta}| \le (1-|\zeta|^2)+|\zeta||\zeta-z| \le 2\delta(\zeta)+|z-\zeta| \le 9|z-\zeta|$, since $1-z\overline{\zeta} = 1-|\zeta|^2+\overline{\zeta}(\zeta-z)$. Hence $|S_N(z,\zeta)| \le 9\cdot 2^N \delta(\zeta)^N |1-z\overline{\zeta}|^{-N-1}$. By Tonelli's theorem and Lemma 6, the far part is

$$\lesssim \int_{\mathbb{D}} |v(\zeta)|\ell(\zeta)\widehat{\omega}(|\zeta|)\,\mathrm{d}A(\zeta) \le \int_{\mathbb{D}} |g(\zeta)|\widetilde{\omega}(\zeta)\,\mathrm{d}A(\zeta) = \|g\|_{A^1_{\widetilde{\omega}}}.$$

In the near region we integrate in ζ first, with z fixed, and we do not use the weight there. If $|z-\zeta| < \delta(\zeta)/4$, then $\delta(\zeta) < \delta(z) + \delta(\zeta)/4$, so that $|z-\zeta| < \delta(z)/3$ and $\delta(\zeta) > 2\delta(z)/3$. Since $\delta\ell$ increases with δ, this gives $\delta(\zeta)\ell(\zeta) \ge (2/3)\delta(z)\ell(z)$. Let

$$A_g(z) = \frac{1}{\delta(z)^2} \int_{|u-z|<\delta(z)/2} |g(u)|\,\mathrm{d}A(u).$$

If $|\zeta - z| < \delta(z)/3$, the disc of centre ζ and radius $\delta(z)/6$ lies in $\{|u-z| < \delta(z)/2\}$, and the subharmonicity of $|g|$ gives $|g(\zeta)| \le 36 A_g(z)$. Using (8) and $\int_{|\zeta-z|<\delta(z)/3} |z-\zeta|^{-1}\,\mathrm{d}A(\zeta) = 2\delta(z)/3$, we find that the inner integral over the near region is at most

$$2^N \cdot \frac{3}{2\delta(z)\ell(z)} \cdot 36 A_g(z) \cdot \frac{2\delta(z)}{3} = \frac{36\cdot 2^N A_g(z)}{\ell(z)}.$$

The factor $\ell(z)$ cancels, and it remains to bound $\int_{\mathbb{D}} A_g\omega\,\mathrm{d}A$. By Tonelli's theorem,

$$\int_{\mathbb{D}} A_g\omega\,\mathrm{d}A = \int_{\mathbb{D}} |g(u)| \int_{|u-z|<\delta(z)/2} \frac{\omega(z)}{\delta(z)^2}\,\mathrm{d}A(z)\,\mathrm{d}A(u).$$

If $|u-z| < \delta(z)/2$, then $\delta(z)/2 < \delta(u) < 3\delta(z)/2$, and so $\delta(z) > 2\delta(u)/3$ and $|u-z| < \delta(u)$. The disc $|z-u| < \delta(u)$ lies in a polar box as in Lemma 1 (iii) with $t = 2\delta(u)$, $c = \pi$ and $\theta = \arg u$: its points satisfy $1-|z| < 2\delta(u)$, and their arguments differ from that of u by at most $\arcsin(\delta(u)/|u|) \le \pi\delta(u)$ when $|u| \ge 1/2$, and by at most $\pi < 2\pi\delta(u)$ otherwise. By Lemma 1 (iii) and (2), its ω-mass is $\lesssim \delta(u)\widehat{\omega}(|u|)$. Hence the inner integral is $\lesssim \widetilde{\omega}(u)$, and the near part is $\lesssim \|g\|_{A^1_{\widetilde{\omega}}}$.

Combining the two parts and applying Lemma 2, we obtain the assertion. □

The near-diagonal argument integrates in ζ before the weight is used. This matters because ω itself need not be regular: a weight in $\mathcal{D}$ may vanish on

intervals or have narrow radial peaks, and only its tails are controlled. The weight enters only through the disc masses in Lemma 1 (iii).

Finally, we check that the kernel produces a solution of the $\overline{\partial}$-equation, and that the Hankel operator is its projection onto the orthogonal complement of A^2_ω.

Lemma 8. *Suppose that $\omega \in \mathcal{D}$, that N is a positive integer, that G is a C^1 function in a neighbourhood of the closed unit disc, and that g is a polynomial. Let*

$$u(z) = \int_{\mathbb{D}} S_N(z,\zeta) g(\zeta) \overline{\partial} G(\zeta)\, \mathrm{d}A(\zeta).$$

Then u is bounded, and $(I - P_\omega)(Gg) = (I - P_\omega)u$.

Proof. Let $v = g\overline{\partial}G$, which is bounded on $\mathbb{D}$. By (8), $|u(z)| \le 2^N \|v\|_\infty \int_{\mathbb{D}} |z-\zeta|^{-1}\, \mathrm{d}A(\zeta) \le 2^{N+2}\|v\|_\infty$, since $\mathbb{D}$ lies in the disc of centre z and radius 2. We write $S_N(z,\zeta) = (z-\zeta)^{-1} + Q(z,\zeta)$, where

$$Q(z,\zeta) = \frac{1}{z-\zeta}\left(\left(\frac{1-|\zeta|^2}{1-z\overline{\zeta}}\right)^N - 1\right).$$

For fixed ζ, the bracket is analytic in z on $\mathbb{D}$ and vanishes at $z = \zeta$, so $Q(\cdot,\zeta)$ is analytic in $\mathbb{D}$. Moreover Q is bounded on $\{|z| \le \varrho\} \times \mathbb{D}$ for each $\varrho < 1$: where $|z-\zeta| \ge (1-\varrho)/2$ this follows from the bound 2^N for the bracket, and otherwise $|\zeta| \le (1+\varrho)/2$ and Q is a difference quotient of a function whose z-derivative is bounded there. Hence $\int_{\mathbb{D}} Q(z,\zeta) v(\zeta)\, \mathrm{d}A(\zeta)$ is analytic in z, by Morera's and Fubini's theorems.

The remaining term is the Cauchy transform of v, which is $Cv(z) = \int_{\mathbb{D}} v(\zeta)(z-\zeta)^{-1}\, \mathrm{d}A(\zeta)$. Let Φ be a smooth function with compact support in $\mathbb{D}$. In our normalization of $\mathrm{d}A$, the Cauchy–Pompeiu formula reads

$$\Phi(\zeta) = \int_{\mathbb{C}} \frac{\overline{\partial}\Phi(z)}{\zeta - z}\, \mathrm{d}A(z),$$

and Fubini's theorem gives

$$-\int_{\mathbb{D}} Cv\, \overline{\partial}\Phi\, \mathrm{d}A = \int_{\mathbb{D}} v(\zeta) \int_{\mathbb{C}} \frac{\overline{\partial}\Phi(z)}{\zeta - z}\, \mathrm{d}A(z)\, \mathrm{d}A(\zeta) = \int_{\mathbb{D}} v\Phi\, \mathrm{d}A.$$

Hence $\overline{\partial}Cv = v$, and therefore $\overline{\partial}u = v = \overline{\partial}(Gg)$, in the sense of distributions on $\mathbb{D}$. By Weyl's lemma for $\overline{\partial}$, the bounded function $Gg - u$ coincides almost everywhere with an analytic function k, which is bounded and hence belongs to A^2_ω. Since $P_\omega k = k$, we get $(I - P_\omega)(Gg) = (I - P_\omega)u$. □

5. Proofs of Theorem 1 and Corollary 1

Proof of Theorem 1. By Hölder's inequality, $f \in L^1_\omega$ and $\ell\mathrm{MO}^1_{\omega,r}(f) \le \ell\mathrm{MO}^p_{\omega,r}(f)$, so we may assume that $p = 1$. Let $M = \|f\|_{\mathrm{BMO}^1_{\omega,r,\log}}$ and $\varrho = \max(r, r_0 + 3)$. By Lemma 4, $\sup_z \ell(z)\mathrm{MO}^1_{\omega,\varrho}(f)(z) \lesssim M$. Let $f = F + h$ be the decomposition of Lemma 5, and fix an integer $N > b$.

Step 1: the operator is well defined. Integrating (6) along the radius from 0 to z gives

(9)
$$|F(z)| \leq |F(0)| + CM \int_0^{|z|} \frac{\mathrm{d}s}{(1-s)\log(e/(1-s))} = |F(0)| + CM \log \ell(z).$$

By Lemma 1 (i) with $r = 0$, $\ell^2 \omega \, \mathrm{d}A$ is a finite measure, and so $F \in L^2_\omega$ and $\int \ell |F| \omega \, \mathrm{d}A < \infty$. Let $g \in H^\infty$. Then $\int \ell |Fg| \omega \, \mathrm{d}A < \infty$, and $\int \ell |hg| \omega \, \mathrm{d}A < \infty$ by (7). Hence $\int \ell |fg| \omega \, \mathrm{d}A < \infty$, and the integral defining $P_\omega(fg)$ converges absolutely by Lemma 3. We have $H^\omega_f g = H^\omega_F g + H^\omega_h g$, where each term is defined by the same formula. If $f \in L^2_\omega$, then $fg \in L^2_\omega$, and the integral is the orthogonal projection.

Step 2: the remainder. By Lemma 3 and (7), and since $\ell \geq 1$,

$$\|H^\omega_h g\|_{L^1_\omega} \leq \|hg\|_{L^1_\omega} + \|P_\omega(hg)\|_{L^1_\omega} \lesssim \int_{\mathbb{D}} \ell |hg| \omega \, \mathrm{d}A \lesssim M \|g\|_{A^1_\omega}.$$

Step 3: the smooth part, for polynomials. Let g be a polynomial. For $0 < t < 1$ let $F_t(z) = F(tz)$, which is smooth in a neighbourhood of the closed disc. Since $\delta(t\zeta)\ell(t\zeta) \geq \delta(\zeta)\ell(\zeta)$, (6) gives $|\overline{\partial} F_t(\zeta)| \leq C_1 M / (\delta(\zeta)\ell(\zeta))$ with C_1 independent of t. Let u_t be the function of Lemma 8 with $G = F_t$. By Lemmas 3 and 8 and Lemma 7 applied to $v = g\overline{\partial} F_t / (C_1 M)$,

$$\begin{aligned} \|H^\omega_{F_t} g\|_{L^1_\omega} = \|(I - P_\omega) u_t\|_{L^1_\omega} &\lesssim \int_{\mathbb{D}} \ell |u_t| \omega \, \mathrm{d}A \\ &\leq \int_{\mathbb{D}} \ell(z) \omega(z) \int_{\mathbb{D}} |S_N(z,\zeta) g(\zeta) \overline{\partial} F_t(\zeta)| \, \mathrm{d}A(\zeta) \, \mathrm{d}A(z) \\ &\lesssim M \|g\|_{A^1_\omega}. \end{aligned}$$

As $t \to 1^-$, $F_t \to F$ pointwise, and $|F_t(z)| \leq |F(0)| + CM \log \ell(z)$ by (9), since $\ell(tz) \leq \ell(z)$. By dominated convergence, $F_t g \to Fg$ in L^2_ω. Since $I - P_\omega$ is bounded on L^2_ω, and L^2_ω convergence implies L^1_ω convergence, we conclude that $\|H^\omega_F g\|_{L^1_\omega} \lesssim M \|g\|_{A^1_\omega}$.

Step 4: the smooth part, for bounded g. Let $g \in H^\infty$ and $g_t(z) = g(tz)$ for $0 < t < 1$. The Taylor polynomials of g_t converge to g_t uniformly on $\mathbb{D}$, so F times them converges to Fg_t in L^2_ω, and Step 3 gives $\|H^\omega_F g_t\|_{L^1_\omega} \lesssim M \|g_t\|_{A^1_\omega} \leq M \|g\|_{A^1_\omega}$. The last inequality holds because the integral means of $|g|$ over circles are nondecreasing. As $t \to 1^-$, $g_t \to g$ boundedly and pointwise, so $Fg_t \to Fg$ in L^2_ω, and the bound passes to g.

Combining Steps 2 and 4, we get $\|H^\omega_f g\|_{L^1_\omega} \lesssim M \|g\|_{A^1_\omega}$ for every $g \in H^\infty$. Polynomials are dense in A^1_ω, since $g_t \to g$ in A^1_ω by dominated convergence applied to the integral means, and so H^ω_f extends to a bounded operator from A^1_ω to L^1_ω. □

Proof of Corollary 1. Let $f \in \mathcal{LB}$ and fix $r > r_0$. Let $z \in \mathbb{D}$ and $\zeta \in D(z,r)$, and let γ be the geodesic from z to ζ, which lies in $D(z,r)$. The Bergman metric has length element $|\mathrm{d}w|/(1-|w|^2)$, and $\ell \gtrsim \ell(z)$ on $D(z,r)$ by (4).

Hence

$$|f(\zeta)-f(z)|\le\int_\gamma|f'(w)||\mathrm{d}w|\le\|f\|_{\mathcal{LB}}\int_\gamma\frac{|\mathrm{d}w|}{(1-|w|^2)\ell(w)}\lesssim\frac{\|f\|_{\mathcal{LB}}\,r}{\ell(z)}.$$

Since replacing the mean by the constant $\overline{f(z)}$ at most doubles the oscillation, $\mathrm{MO}^1_{\omega,r}(\overline{f})(z)\lesssim\|f\|_{\mathcal{LB}}/\ell(z)$. Integrating $|f'|$ along the radius from 0 to ζ instead, as in (9), gives $|f(\zeta)|\le|f(0)|+\|f\|_{\mathcal{LB}}\log\ell(\zeta)$, so that $f\in L^2_\omega$ by Lemma 1 (i). Hence $\overline{f}\in\mathrm{BMO}^1_{\omega,r,\log}\cap L^2_\omega$, and Theorem 1 applies. □

6. Concluding remarks

Where the logarithm is used. By [4, Theorem 1(ii)], $\|B^\omega_\zeta\|_{A^1_\omega}$ grows exactly like $\log(1/(1-|\zeta|))$, so Lemma 3 cannot be improved for functions u concentrated near a point. In the proof, this loss is paid by the factor $1/\ell$ in the oscillation of f: once in the gradient bound (6) for the smooth part, and once in the residual estimate (7). This is why a logarithmic condition, rather than plain bounded mean oscillation, appears at the endpoint $A^1_\omega\to L^1_\omega$.

The radius. The space $\mathrm{BMO}^p_{\omega,r,\log}$ is defined in [2] for $r\ge r_0$, and stated there to be independent of r. For $r_0<s<r$ we have $\mathrm{BMO}^1_{\omega,s,\log}\subset\mathrm{BMO}^1_{\omega,r,\log}$ by Lemma 4. The reverse inclusion holds since $\mathrm{MO}^1_{\omega,s}(f)(z)\le 2\,(\omega(D(z,r))/\omega(D(z,s)))\mathrm{MO}^1_{\omega,r}(f)(z)$, and the ratio of masses is bounded by (1). The same arguments apply to $\mathrm{MO}^p_{\omega,r}$. Hence the spaces $\mathrm{BMO}^p_{\omega,r,\log}$, $r>r_0$, all coincide with equivalent norms. Our argument uses $r>r_0$ only in Lemma 4, where a disc of radius r_0 must fit inside two neighbouring discs of radius r. If the fixed radius in the definition of $\mathrm{BMO}^p_{\omega,\log}$ is $r=r_0$ itself, Theorem 1 applies provided $\mathrm{BMO}^p_{\omega,r_0,\log}\subset\mathrm{BMO}^p_{\omega,r,\log}$ for some $r>r_0$, which is part of the independence stated in [2, Section 3.4]. Since the upper bound in (1) holds for every $r>0$ by Lemma 1 (iii), and the lower bound persists when r increases, (1) holds for every r larger than an admissible r_0. So Theorem 1 applies to every radius r for which (1) holds, except possibly the smallest such radius.

The classical case. For the standard weights $\omega(z)=(\alpha+1)(1-|z|^2)^\alpha$ one may take $b=a=\alpha+1$, and (1) holds for every $r>0$, so that r_0 can be any positive number and Theorem 1 holds for every $r>0$. In this case P_ω has the explicit kernel $(1-\overline{z}\zeta)^{-2-\alpha}$, but the proof above does not use it. By Lemma 8, any solution of the $\overline{\partial}$-equation with enough decay, followed by the projection $I-P_\omega$ that is already part of the definition of H^ω_f, gives the canonical solution. This is what allows a general $\omega\in\mathcal{D}$, for which no closed form of B^ω_z is available.

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